

Ying Yuan

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EDUCATION

Tsinghua University, China

Sept 2020 - Now

BEng Computer Science and Technology, Institute for Interdisciplinary Information Sciences (Yao Class)

GPA: 3.95/4.00, Ranking: 5/76

INTERNSHIPS

Reinforcement Learning for Dexterous Manipulation

Feb 2023 - Aug 2023

Research Intern; Advisor: Xiaolong Wang, Assistant Professor, ECE Department, UC San Diego

San Diego, United States

- Proposed Robot Synesthesia, a novel point cloud-based tactile representation inspired by human tactile-visual synesthesia.
- Built a system for in-hand dexterous manipulation utilizing visuotactile sensing to solve complex tasks in the real world such as double-ball rotation and generalize to novel objects.

Generalization and Adaptive Robotics System

Jun 2022 - Mar 2023

Research Assistant; Advisor: Yi Wu, Assistant Professor, IIIS, Tsinghua University

Beijing, China

- Built block stacking and Franka kitchen environments with a Franka Panda arm based on PyBullet.
- Designed the Phasic Self-Expansion algorithm to tackle a long-horizon sparse-reward robotic manipulation problem, where the agent searches for more challenging goals automatically based on current successful trajectories.

PUBLICATIONS

Robot Synesthesia: In-Hand Manipulation with Visuotactile Sensing.

2024

Ying Yuan*, Haichuan Che*, Yuzhe Qin*, Binghao Huang, Zhao-Heng Yin, Kang-Won Lee, Yi Wu, Soo-Chul Lim, Xiaolong Wang.

Submitted to ICRA 2024, arXiv preprint 2312.01853

Robot Generating Data for Learning Generalizable Visual Robotic Manipulation.

2024

Yunfei Li*, **Ying Yuan***, Wei Fu, Jiaxuan Gao, Zekai Xu, Yi Wu.

Submitted to AAAI 2024

Robot Synesthesia: In-Hand Manipulation with Visuotactile Sensing.

2023

Ying Yuan*, Haichuan Che*, Yuzhe Qin*, Binghao Huang, Zhao-Heng Yin, Kang-Won Lee, Yi Wu, Soo-Chul Lim, Xiaolong Wang.

NeurIPS 2023 Workshop WTP

Spotlight

Self-Generating Data for Goal-Conditioned Compositional Problems.

2023

Ying Yuan, Yunfei Li, Yi Wu.

ICLR 2023 Workshop RRL

Poster

HONORS AND AWARDS

Yao Award (Recognition Prize, for recognizing the outstanding performances of selected junior students)

2023

School-level Scholarship (in total 30,000 CNY)

2021-2023

Scholarship for Tsinghua School Talent Training Program (in total 15,000 CNY)

2021-2023

TECHNICAL SKILLS

Languages: English (Fluent, TOEFL 116, GRE 331+4.0); Chinese (Native)

Programming languages: Python, C++, MATLAB, Verilog

Libraries and tools: Pytorch, TensorFlow, Git, LaTeX, Kubernetes, CUDA, Docker

RELEVANT COURSEWORK

Computer Science: Data Structures in the Real World (A+), Deep Learning (A+), Natural Language Processing (A+), Theory of Computation (A), Machine learning (A), Computer Vision (A), Intelligent Systems and Robotics (A)

Mathematics & Physics: Calculus (A), Linear Algebra (A-), Abstract Algebra (A-), General Physics (A), The Physics of Information (A)